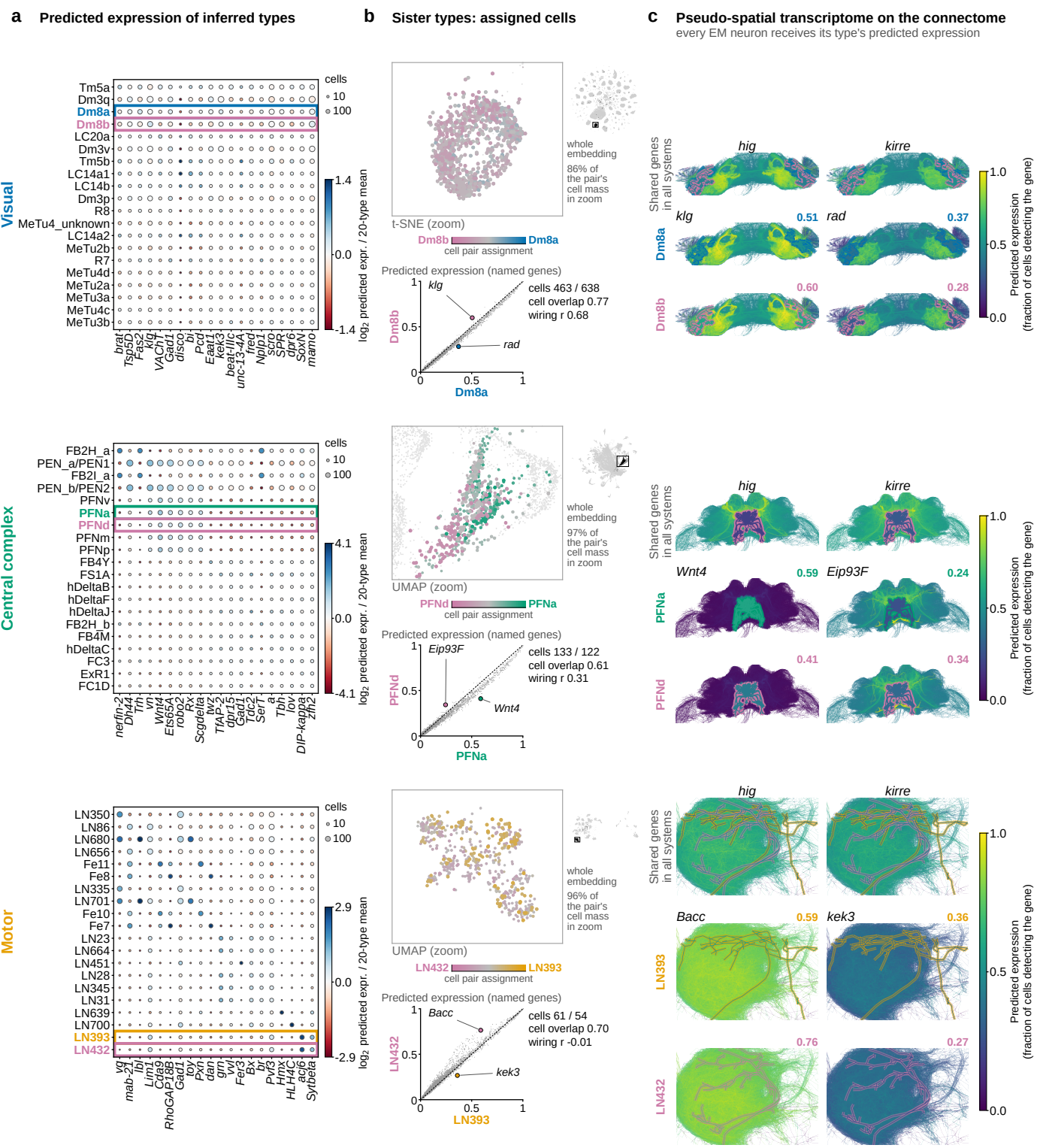

a
Predicted expression of inferred types
b
Sister types: assigned cells
c
Pseudo-spatial transcriptome on the connectome
every EM neuron receives its type's predicted expression
Visual
Central complex
Motor
cells
10
100
log2 predicted expr. / 20-type mean
Tm5a
Dm3q
Dm8a
Dm8b
LC20a
Dm3v
Tm5b
LC14a1
LC14b
Dm3p
R8
MeTu4_unknown
LC14a2
MeTu2b
R7
MeTu4d
MeTu2a
MeTu3a
MeTu4c
MeTu3b
brat
Tsp5D
Fas2
klg
VAChT
Gad1
disco
bi
Pcd
Eaat1
kek3
beat-IIIc
unc-13-4A
fred
Nplp1
scro
SPR
dpr6
SoxN
mamo
t-SNE (zoom)
whole embedding
86% of the pair's cell mass in zoom
cell pair assignment
Predicted expression (named genes)
cells 463 / 638
cell overlap 0.77
wiring r 0.68
Shared genes in all systems
hig
kirre
klg
0.51
rad
0.37
0.60
0.28
Predicted expression (fraction of cells detecting the gene)
FB2H_a
PEN_a/PEN1
FB2I_a
PEN_b/PEN2
PFNv
PFNa
PFNd
PFNm
PFNp
FB4Y
FS1A
hDeltaB
hDeltaF
hDeltaJ
FB2H_b
FB4M
hDeltaC
FC3
ExR1
FC1D
nerfin-2
Dh44
Trh
vn
Wnt4
Ets65A
robo2
Rx
Scgdelta
Pyr2
TfAP-2
dpr15
Gad1
Tdc2
SerT
a
Tbh
lov
DIP-kappa
Zfh2
UMAP (zoom)
cells 133 / 122
cell overlap 0.61
wiring r 0.31
Wnt4
0.59
Eip93F
0.24
0.41
0.34
LN350
LN86
LN680
LN656
Fe11
Fe8
LN335
LN701
Fe10
Fe7
LN23
LN664
LN451
LN28
LN345
LN31
LN639
LN700
LN393
LN432
vg
mab-21
lbl
Lim1
Cda9
RhoGAP18B
Gad1
toy
Pan
dan
wit
Fer3
Bx
bi
Pvf3
Hmx
HLH4C
acj6
Sytbeta
cells 61 / 54
cell overlap 0.70
wiring r -0.01
Bacc
0.59
kek3
0.36
0.76
0.27